

React Fundamentals

tsevdos.me / [@tsevdos](https://twitter.com/tsevdos)

Agenda

All the content can be found [here](#).

- what is react
- core principles
- JSX
- components

Rules

Feel free to interrupt me for:

- questions
- relevant comments

What is React

React is a library for building user interfaces.

- virtual DOM
- JSX
- event handling
- performance

Core principles

- composition
- declarative
- unidirectional dataflow
- explicit mutations

Composition

- divide and conquer
- hide complexity
- comes from functional programming

Composition

Composition

```
<Widget>  
  <SearchForm />  
  <Results>  
    <Header />  
    <SportsTable />  
    <ElectronicsTable />  
  </Results>  
</Widget>
```


Avatar sample code

```
function getProfilePhoto(username) {  
    return "https://twitter.com/photos/" + username;  
}  
function getProfileLink(username) {  
    return "https://twitter.com/" + username;  
}  
  
function getAvatar(username) {  
    return {  
        photo: getProfilePhoto(username),  
        link: getProfileLink(username),  
    };  
}  
  
getAvatar("tseydos");
```

Avatar (React code)

```
const ProfilePhoto = (props) {  
  return <img src={"https://twitter.com/photos/" + props.username} />;  
}  
  
function ProfileLink(props) {  
  return (  
    <a href={"https://twitter.com/" + props.username}>  
      { props.username }  
    </a>  
  );  
}  
...
```

Avatar (React code)

```
...  
function Avatar(props) {  
  return (  
    <div>  
      <ProfilePhoto username={props.username} />  
      <ProfileLink username={props.username} />  
    </div>  
  );  
}  
  
<Avatar username="tsevdos" />
```

Imperative and Declarative

- imperative programming is a programming paradigm that uses statements that change a program's state
- declarative programming is a programming paradigm that expresses the logic of a computation without describing its control flow

Imperative (How)

```
const numbers = [1, 2, 3, 4, 5];  
  
let total = 0;  
for (let i = 0; i < numbers.length; i++) {  
    total += numbers[i];  
}
```

Declarative (What)

```
const numbers = [1, 2, 3, 4, 5];  
const total = numbers.reduce((total, item) => {  
  return total + item;  
}, 0);
```

JavaScript built in methods

- map
- reduce
- filter
- find

Declarative

- reduce side effects and mutability
- more clear / readable code
- less errors / bugs

React is declarative

```
$("#btn").click(function () {  
  $(this).toggleClass("active");  
  if ($(this).text() === "Active") {  
    $(this).text("Inactive");  
  } else {  
    $(this).text("Active");  
  }  
});
```

```
<Button onClick="handleClick" isActive={isActive} />  
  
setIsActive(!isActive);
```

Unidirectional dataflow

Explicit mutations

```
const [name, setName] = useState("");  
  
setName("John");
```

Rendering elements

- `React.createElement`
- `JSX`
- `virtual DOM`

DOM scripting:

document.createElement

```
// JSX/00.html
const rootElement = document.getElementById("app");
const element = document.createElement("div");

element.textContent = "Hello World";
element.className = "container";
rootElement.appendChild(element);
```

React.createElement

```
// JSX/01.html
const app = document.getElementById("app");
const root = createRoot(app);
const element = React.createElement(
  "div",
  { className: "container" },
  "Hello World"
);

root.render(element);
```

React.createElement

```
React.createElement(type, [props], [...children]);
```

React.createElement

```
// JSX/02.html
const app = document.getElementById("app");
const root = createRoot(app);

const element = React.createElement(
  "div",
  { className: "container" },
  React.createElement("div", null, "Div 1"),
  React.createElement(
    "div",
    null,
    React.createElement("h2", null, "Title"),
    React.createElement("p", null, "Paragraph inside div 2")
  )
);
```


Virtual DOM

The virtual DOM (VDOM) is an in-memory representation of real DOM. The representation of a UI is kept in memory and synced with the “real” DOM. It’s a step that happens between the render function being called and the displaying of elements on the screen. This entire process is called reconciliation.

Virtual DOM

JSX

```
// JSX/03.html
// const element = React.createElement(
//   "div",
//   { className: "container" },
//   "Hello World"
// );

const element = <div className="container">Hello World</div>;
```

JSX

```
// JSX/04.html
const element = (
  <div className="container">
    <div>Div 1</div>
    <div>
      <h2>Title</h2>
      <p>Paragraph inside div 2</p>
    </div>
  </div>
);
```

JSX interpolation

```
// JSX/05.html
const title = "Hello World";
const myClassName = "container";

const element = <div className={` ${myClassName}-1`} >{title}</div>;
```

Babel transpilation / compilation

- example

Components

- what is a component
- component props
- component children
- conditional rendering

Components

A component is a function (~~or a class~~) which optionally accepts input and returns a React element (or null).

Still JSX (no components)

```
// components/00.html
const element = (
  <div className="container">
    <div>Hello World</div>
    <div>Hello World</div>
  </div>
);
```

Still JSX (no components)

```
// components/01.html
const myDiv = <div>Hello World</div>;
const element = (
  <div className="container">
    {myDiv}
    {myDiv}
  </div>
);
```

Our first functional reusable component

```
// components/02.html
const MyDiv = (props) => {
  return <div>{props.msg}</div>;
};

const element = (
  <div className="container">
    <MyDiv msg="Hello World" />
    <MyDiv msg="Hi all!!!" />
  </div>
);
```

Component rules

User-defined components must be capitalized in JSX (lower-case tag names are considered to be HTML tags).

- `<mydiv />` compiles to `React.createElement('mydiv')`
(html tag)
- `<Mydiv />` compiles to `React.createElement(Mydiv)`

Functional component transpilation

- Babel example

Components and children

```
// components/03.html
const MyDiv = (props) => {
  return <div>{props.children}</div>;
};

const element = (
  <div className="container">
    <MyDiv>Hello World</MyDiv>
    <MyDiv>
      Welcome all!
      <MyDiv>Hi I'm a component</MyDiv>
    </MyDiv>
    <MyDiv>
      <h1>Title</h1>
      <p>Welcome</p>
    </MyDiv>
  </div>
);
```

Children

`Props.children` displays whatever you include between the opening and closing tags when invoking a component.

- freedom and composition
- almost everything can be a child (element, component and function)

Valid props

- string
- number
- boolean
- array
- object
- function
- symbol

Components

```
// components/04.html
const Avatar = (props) => {
  return (
    <div>
      <h3>{props.username}</h3>
      <img width="100" src={props.imageUrl} />
      <p>My age is {props.age}</p>
      <p>My hobbies are:</p>
      <ul>
        {props.hobbies.map((hobbie) => (
          <li key={hobbie}>{hobbie}</li>
        ))}
      </ul>
    </div>
  )
}
```

Components

```
// components/05.html  
<Widget  
  title="Website traffic"  
  logo="https://cdn-icons-png.flaticon.com/512/5610/5610944.png"  
  data={data}  
>
```

Components

```
// components/06.html
<div>
  <Widget
    title="Website traffic"
    logo="https://cdn-icons-png.flaticon.com/512/5610/5610944.png"
    data={data}
  />
  <Widget
    title="Website errors"
    logo="https://cdn-icons-png.flaticon.com/512/753/753344.png"
    data={data2}
  />
</div>
```

Components

```
// components/07.html
const Widget = (props) => {
  return (
    <div>
      <h2>{props.title}</h2>
      <img width="30" height="30" src={props.logo} />
      {props.data.map((entry) => (
        <WidgetEntryItem key={entry.title} {...entry} />
      ))}
    </div>
  );
};
```

Conditional rendering: If/Else

```
// components/08.html
const User = ({ username }) => {
  if (username) {
    return <div>Hello, {username}</div>;
  }

  return <div>Hi stranger!</div>;
};
```

Conditional rendering: Ternary operator

```
// components/09.html
const User = ({ username }) => {
  return (
    <div>
      {username ? <>Hello, {username}</> : <>Hi stranger!</>}
    </div>
  );
};
```

Conditional rendering: Ternary operator

```
// components/10.html
const User = ({ username }) => {
  return (
    <div>
      {username ? `Hello, ${username}` : "Hi stranger!"}
    </div>
  );
};
```

Conditional rendering:

Short-circuit operator (&&)

```
// components/11.html
const FavoriteColorsList = ({ list }) => {
  return (
    <div>
      {list.length > 0 && (
        <div>
          {list.map((color) => (
            <span key={color}>{color}, &nbsp;</span>
          ))}
        </div>
      )}
    </div>
  )
}
```


Conditional rendering: Element variables

```
// example components/12.html
const User = ({ isLoggedIn }) => {
  let button;

  if (isLoggedIn) {
    button = <button>Logout</button>;
  } else {
    button = <button>Login</button>;
  }

  return <div>{button}</div>;
};
```

Components

- what is a component
- component state
- component hooks
- event handlers

Components

A component is a function (~~or a class~~) which optionally accepts input and returns a React element (or null).

useState hook

useState hook enqueues changes to the component state and tells React that this component and its children need to be re-rendered with the updated state. This is the primary method you use to update the user interface in response to event handlers and server responses.

Component state

```
// state/00.html
const LikeCount = () => {
  const [counter, setCounter] = React.useState(0);
  const handleLike = () => {
    setCounter(counter + 1);
  };

  return (
    <div>
      <div className="emoji">👉 {counter}</div>
      <button onClick={handleLike}>Like!</button>
    </div>
  );
};
```

Use useState hook correctly

- Only update the state with the appropriate function
- State updates are asynchronous (React may batch multiple `setState()` calls into a single update for performance)

Component state

```
// state/01.html
const LikeCount = () => {
  const [counter, setCounter] = React.useState(0);
  const handleLike = () => {
    setCounter((counter) => counter + 1);
  };
  const handleDislike = () => {
    setCounter((counter) => counter - 1);
  };
  ...
}
```

Use `useState` hook correctly

```
// Wrong
counter = 5; // this will not re-render a component
// Correct
const [counter, setCounter] = React.useState(0);
setCounter(5);

// Might cause a problem
setCounter(counter + 1);
// Correct
setCounter((counter) => counter + 1);
```


Do not mutate the state

```
// state/02.html
const changeName = () => {
  const newProfile = profile;
  profile.user.name = "New Name";
  // console.log(newProfile);
  setProfile(newProfile);
};
```

Updating the state correctly

```
// state/03.html

const changeName = () => {
  setProfile((profile) => ({
    ...profile,
    user: {
      ...profile.user,
      name: "New Name",
    },
  })));
};
```

Updating the state correctly

Immutable tricks for arrays

```
// Arrays
// Spread Operator (ES6)
setState([...arr, "new value"]);
// Array.prototype.slice() (ES5)
const newArr = arr.slice();
newArr.push("new value");
setState(newArr);
```

Updating the state correctly

Immutable tricks for objects

```
// Objects  
// Spread Operator (ES6)  
setState({ ...user, name: "New Name" });  
// Object.assign (ES6)  
const newUser = Object.assign({}, user);
```

Using state

The state of one component can be the props of another one.

```
// state/04.html
const Form = () => {
  ...

  return (
    <div>
      <Hello name={name} />
      <input type="text" name="name" onChange={handleOnChange} value={name} />
    </div>
  );
};
```

Components and events

- SyntheticEvent
- cross-browser wrapper around the browser's native event
- it has the same interface as the browser's native event, including `stopPropagation()` and `preventDefault()`
- you have access to the native event using `event.nativeEvent`

Components and events

- react events are named using camelCase, rather than lowercase
- supported events

Styling and CSS

- global CSS stylesheets
- CSS modules
- CSS in JS
- build-time CSS
- inline styling
- UI libraries

Global CSS stylesheets

- global styles
- `styling-and-css/00.html`

CSS classes

- linked styles
- `styling-and-css/01.html`

In-line styles

```
// styling-and-css/02.html

const buttonStyles = {
  color: "#fff",
  background: "linear-gradient(135deg, #3057d5, #4d7cff)",
  border: "1px solid rgba(48, 87, 213, 0.2)",
  ...
};

<a
  href="https://getbootstrap.com"
  className="btn btn-primary btn-lg"
  style={buttonStyles}
>
  My button
</a>
```

React and styling is a huge topic

- global CSS stylesheets
- CSS modules
- CSS in JS
- build-time CSS
- inline styling
- UI libraries

Recap

- what is react
- core principles
- JSX
- components

Recap: Core principles

- composition
- declarative
- unidirectional dataflow
- explicit mutations

Recap: Basics

- `React.createElement`
- JSX
- virtual DOM

Recap: Components

- props
- state
- hooks
- children
- conditional rendering
- event handlers

Recap: Styling and CSS

- global CSS stylesheets
- CSS modules
- CSS in JS
- build-time CSS
- inline styling
- UI libraries

That's all folks

Questions / Discussions?